AIZG565
Machine Learning

BITS Pilani
Pilani Campus

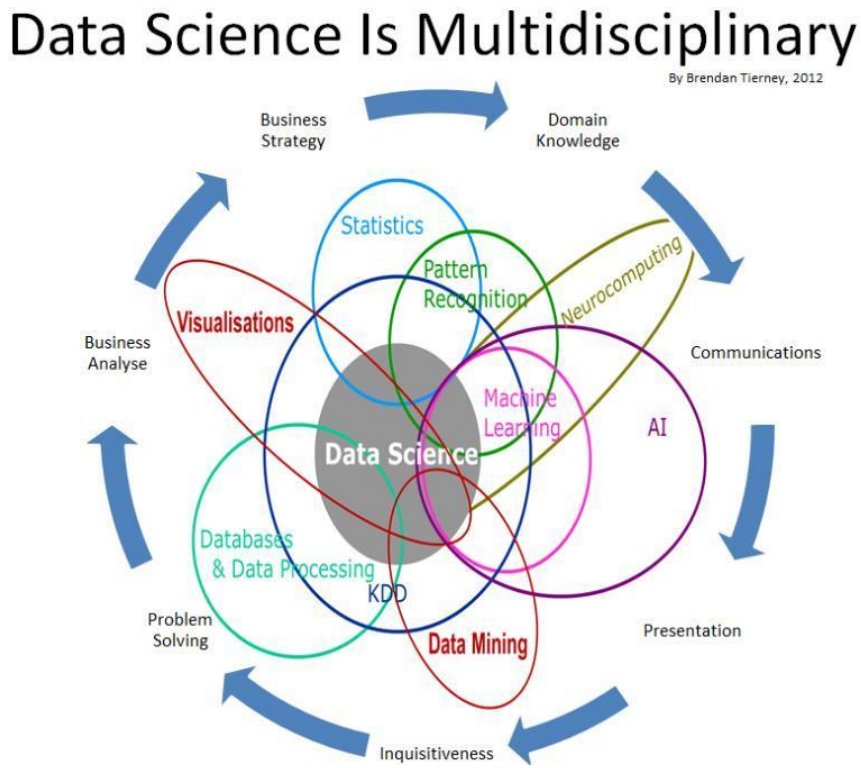
Anand Narasimhamurthy
CSIS, BITS Goa campus

innovate achieve lead

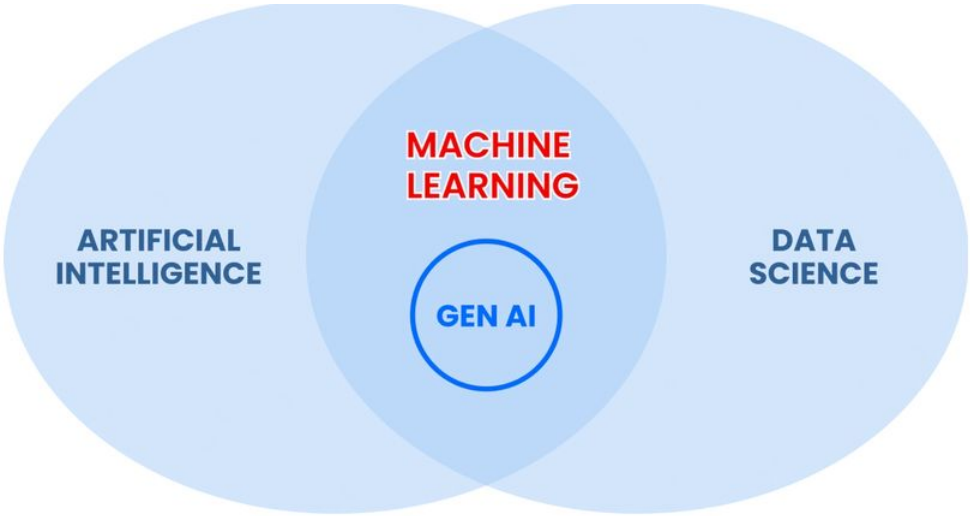
BITS Pilani
Pilani Campus

An overview of Machine Learning and data analytics

Data science/AI/ML etc.



https://www.datasciencecentral.com/difference-of-data-science-machine-learning-and-data-mining/



https://aristeksystems.com/blog/basics-of-ai-data-science-ml-deep-learning-and-generative-ai/

Data ecosystem and lifecycle

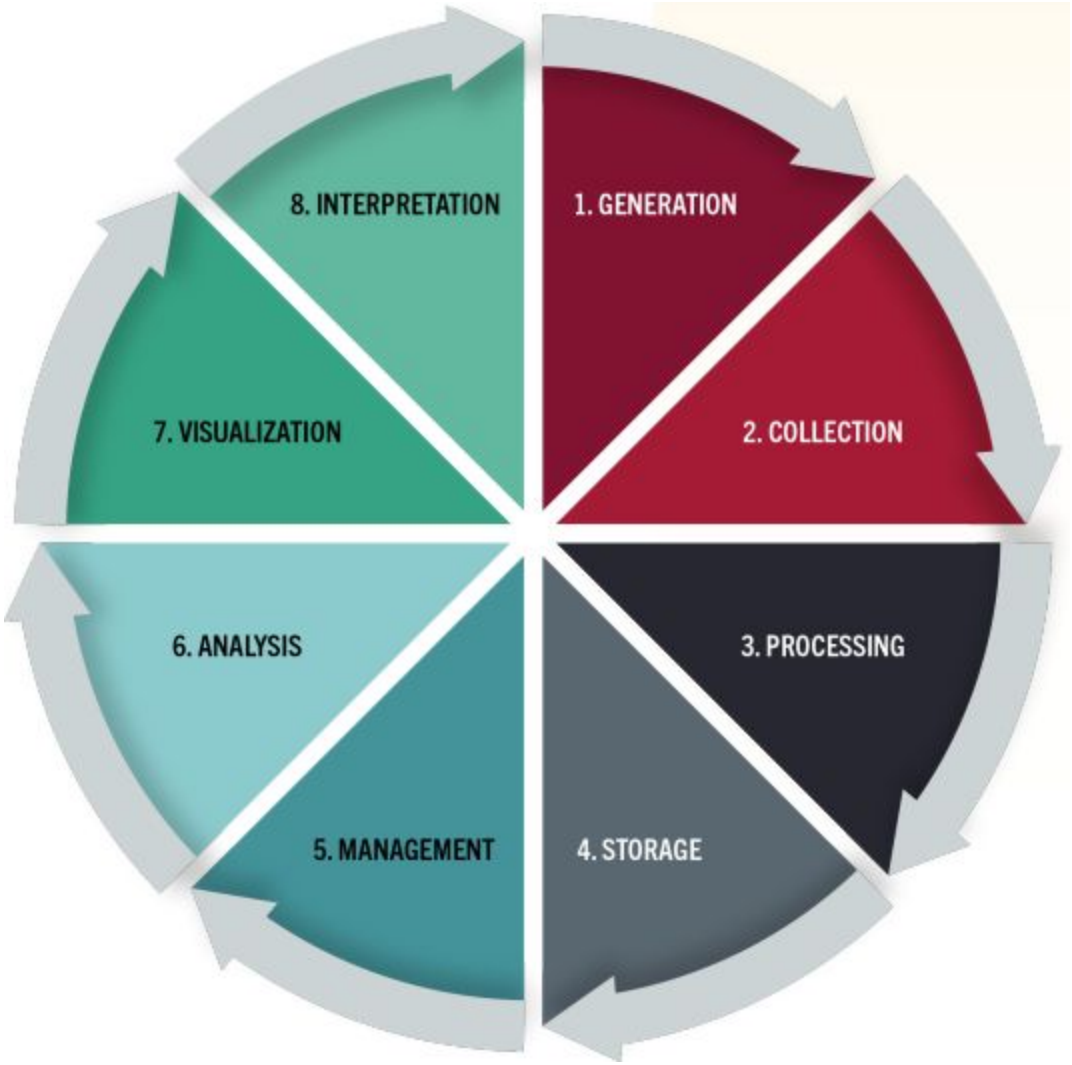


Image reproduced from

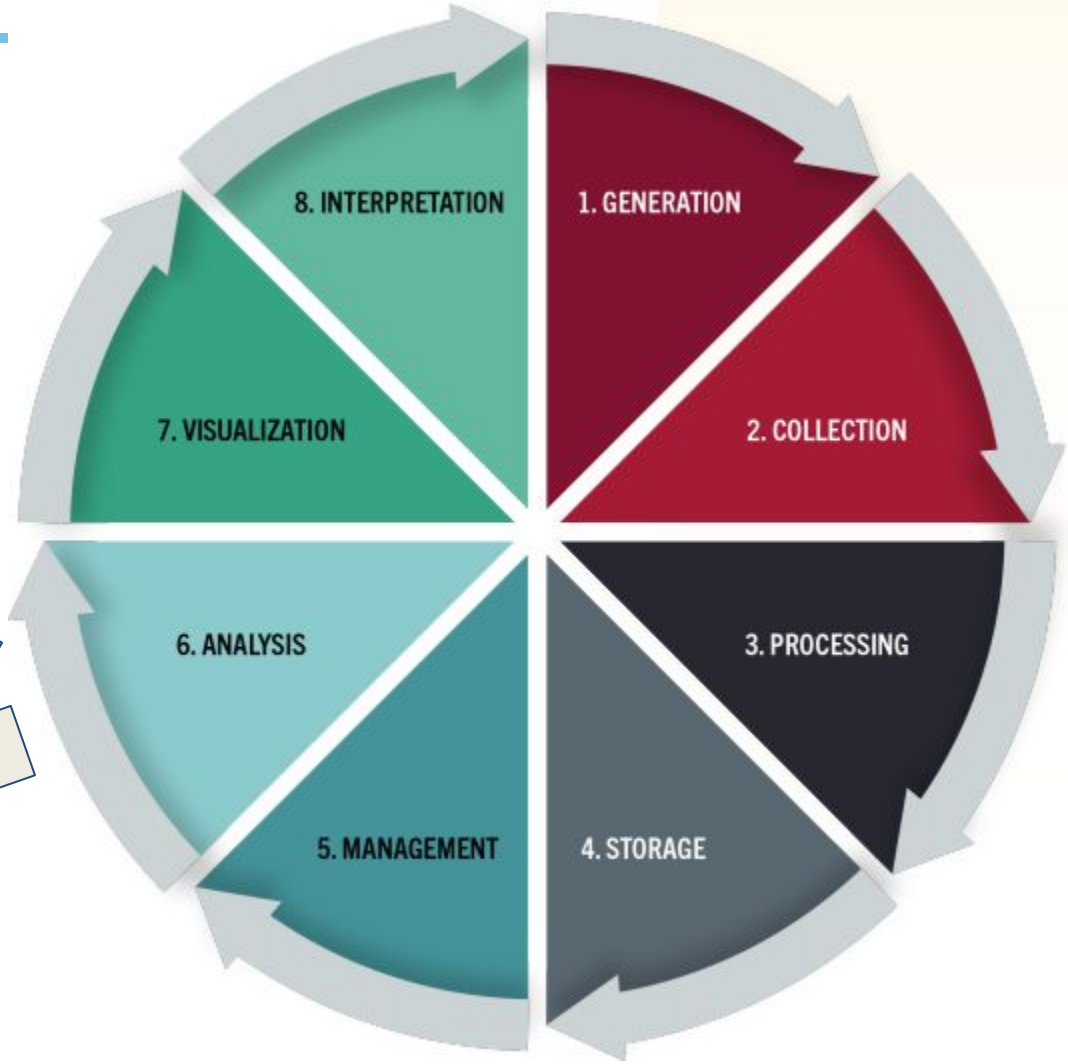
Free Data & Analytics E-Book

Harvard Business School

Types of analytics

Descriptive analytics

Predictive analytics



An overview of common analytics tasks

Common Data Mining tasks

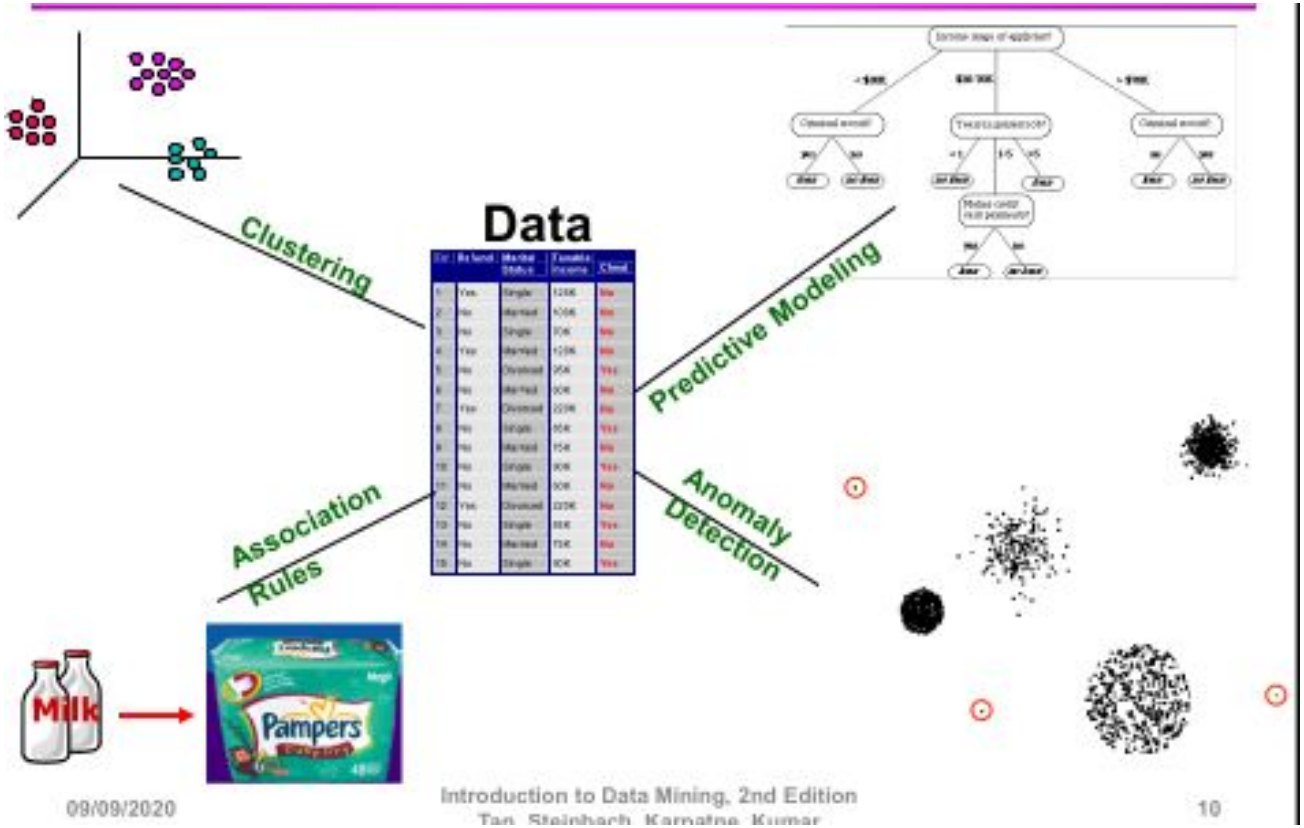
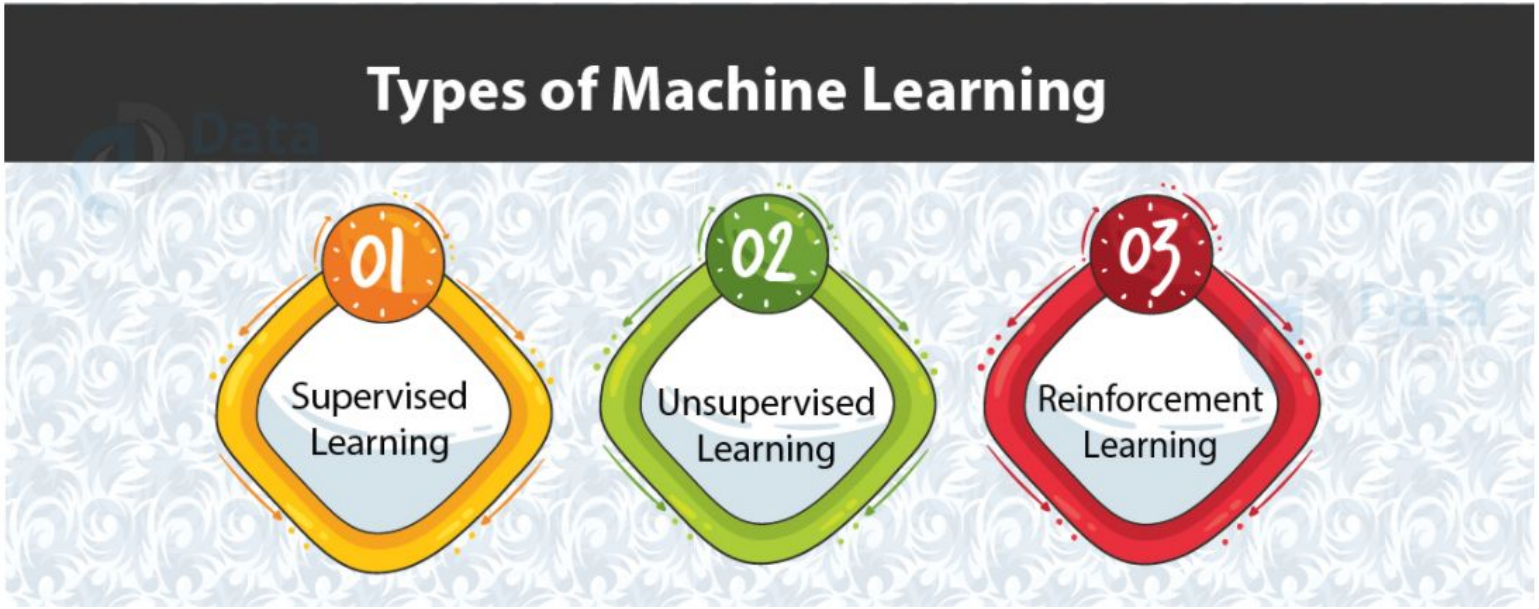


Image source : Companion slides of the text
Introduction to Data Mining, Pan,Steinbach,Kumar

A few types of Machine Learning problems



Source : https://data-flair.training/blogs/machine-learning-infographic/

Supervised and unsupervised learning

In many practical applications there are,

- some easy-to-measure quantities (generally called Xs)
  - Age; Gender; Income; Education level; etc.
- and a difficult-to-measure quantity (generally called the Y)
  - Amount of loan to give; Will she buy or not; How many days will he stay in the hospital; etc.

Supervised and unsupervised learning

- Supervised learning is about computing the Y using the Xs, assuming availability of data samples with Xs and corresponding Ys (usually historical data)
- Unsupervised learning is about computing patterns within easy to measure attributes (the Xs)

Supervised learning overview

Forecasting quarterly sales of a product

Predicting length of patient stay in a hospital

In many practical applications there is a need to predict quantities of interest with reasonable accuracy

Predicting whether a loan applicant is likely to default

Predicting stock prices

Typically these quantities are either difficult to measure or are forecasts.

Common supervised learning subtasks

Categorical
Examples.
- Will the loan applicant default? Yes/No
- Will this customer be a High/Medium/ Low value over next year?

Classification
Eg. Logistic regression
Naive Bayes

Target variable (Y)

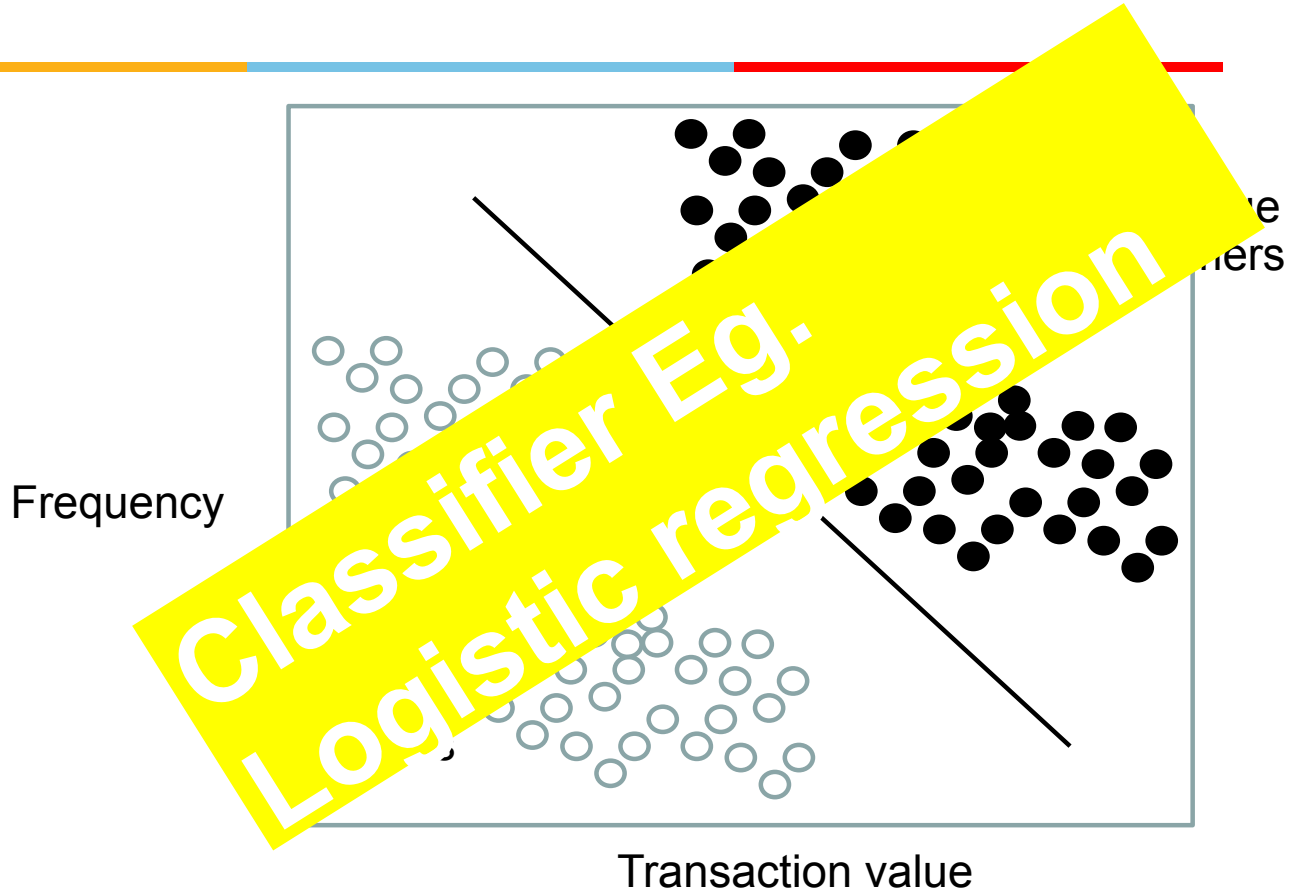
Numeric
Examples
- Quarterly sales of a product
- Length of stay in hospital

Regression
Eg. Linear regression

Some ordering of coordinates eg. wrt time
- Predicting stock price on day t taking into account past stock prices at time t-1, t-2 etc.

Time series

Learning Models: Classification



An illustration of classification with two attributes :

Xs : Transaction value, Frequency

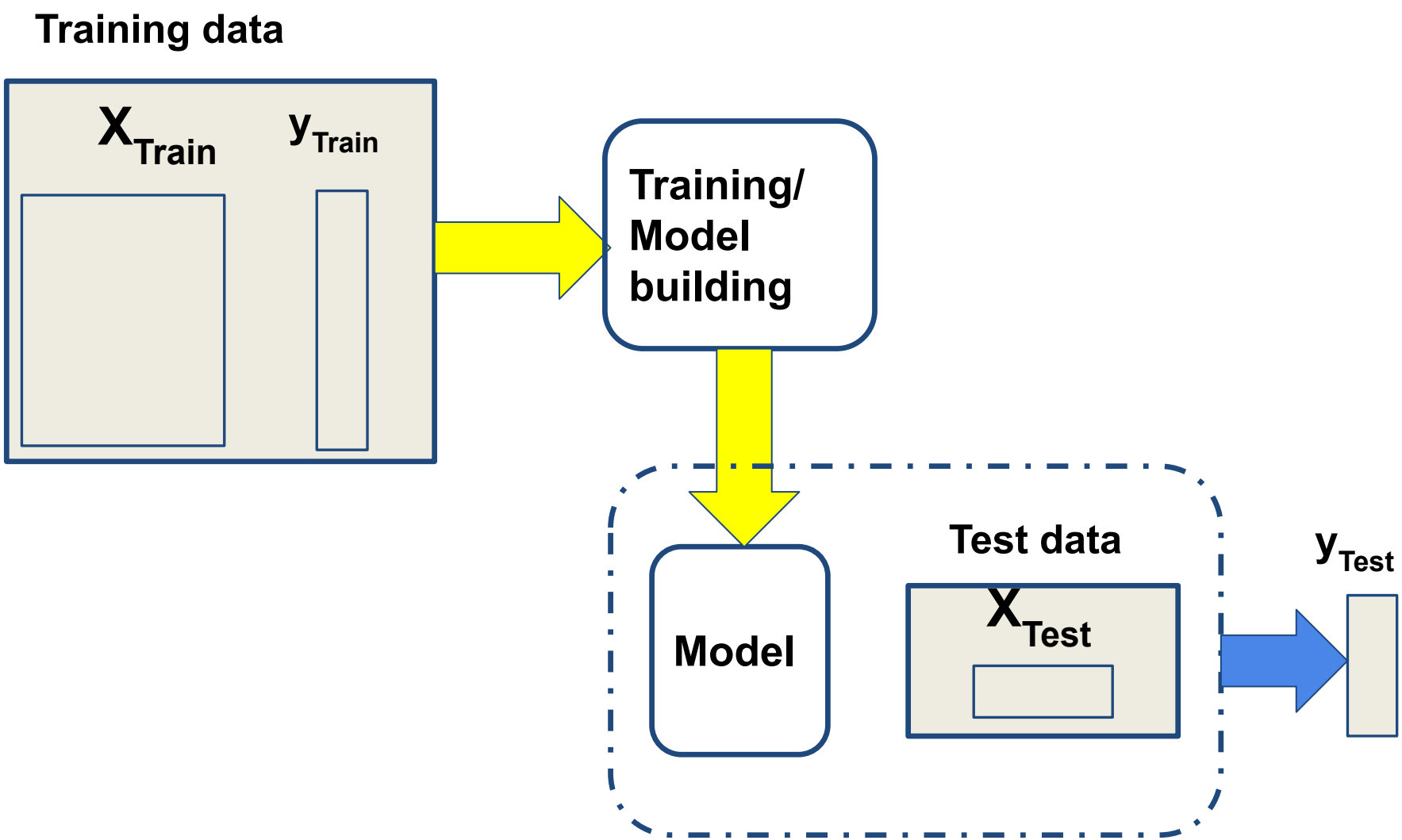
Y (target variable) : High value/Low value customer

Learning Models: Regression

Generally,

- **target variable** (also termed **response variable**) is a **dependent variable** representing something we are interested in predicting (and difficult to measure directly)

- **explanatory variables** (also termed **predictor variables**) are **independent variables** which are “easy” to measure

Suppose in the previous example,
Xs : Transaction value, Frequency
Y (the target variable) : Net worth of the individual

A high level overview of training in supervised learning